

Power BI Case Study: Hospital Performance Analysis

Business Problem:

The goal of this project is to analyze hospital operations, patient data, and financial performance using a Power BI dashboard. This dashboard helps hospital management make data-driven decisions, monitor key metrics, and improve overall efficiency.

Dataset Description:

Appointment ID – Unique identifier for each appointment
Patient ID – Unique identifier for each patient
Doctor ID – Unique identifier for each doctor
Date – Date of appointment
Status – Current status of the appointment
Bill ID – Unique identifier for each bill
Treatment ID – Unique identifier for each treatment
Amount – Amount charged for treatment
Payment Status – Paid, Pending, Partially Paid, or Unpaid
Date Paid – Date when payment was completed
Doctor Name – Name of the doctor
Department – Department of the doctor
Experience – Years of experience of the doctor
Patient Name – Name of the patient
Age – Age of the patient
Gender – Gender of the patient
City – City of the patient
Disease – Patient's medical condition
Treatment Type – Type of treatment provided
Outcome – Result of the treatment
Cost Category – High cost / Low cost classification

Data Preparation (Power Query):

- Remove duplicate records
- Handle missing values
- Remove blank rows
- Standardize text formats
- Fix mixed date formats
- Clean text from numeric fields
- Remove unnecessary columns
- Remove random notes columns
- Correct data types

Data Modeling:

Create relationships between Appointments, Billing, Doctors, Patients, and Treatments tables. Use a star schema model for better performance and efficient querying.

DAX Measures:

Total Patients = `DISTINCTCOUNT(Patients[Patient ID])`
Total Revenue = `SUM(Billing[Amount])`
Total Doctors = `DISTINCTCOUNT(Doctors[Doctor ID])`
Total Appointments = `COUNT(Appointments[Appointment ID])`

Average Billing = AVERAGE(Billing[Amount])

Dashboard Visualizations:

KPI Cards: Total Patients, Total Revenue, Total Doctors, Total Appointments, and Average Billing.

Outcome:

The Power BI dashboard enables management to gain real-time insights, improve decision-making, and effectively monitor hospital operations and financial performance.